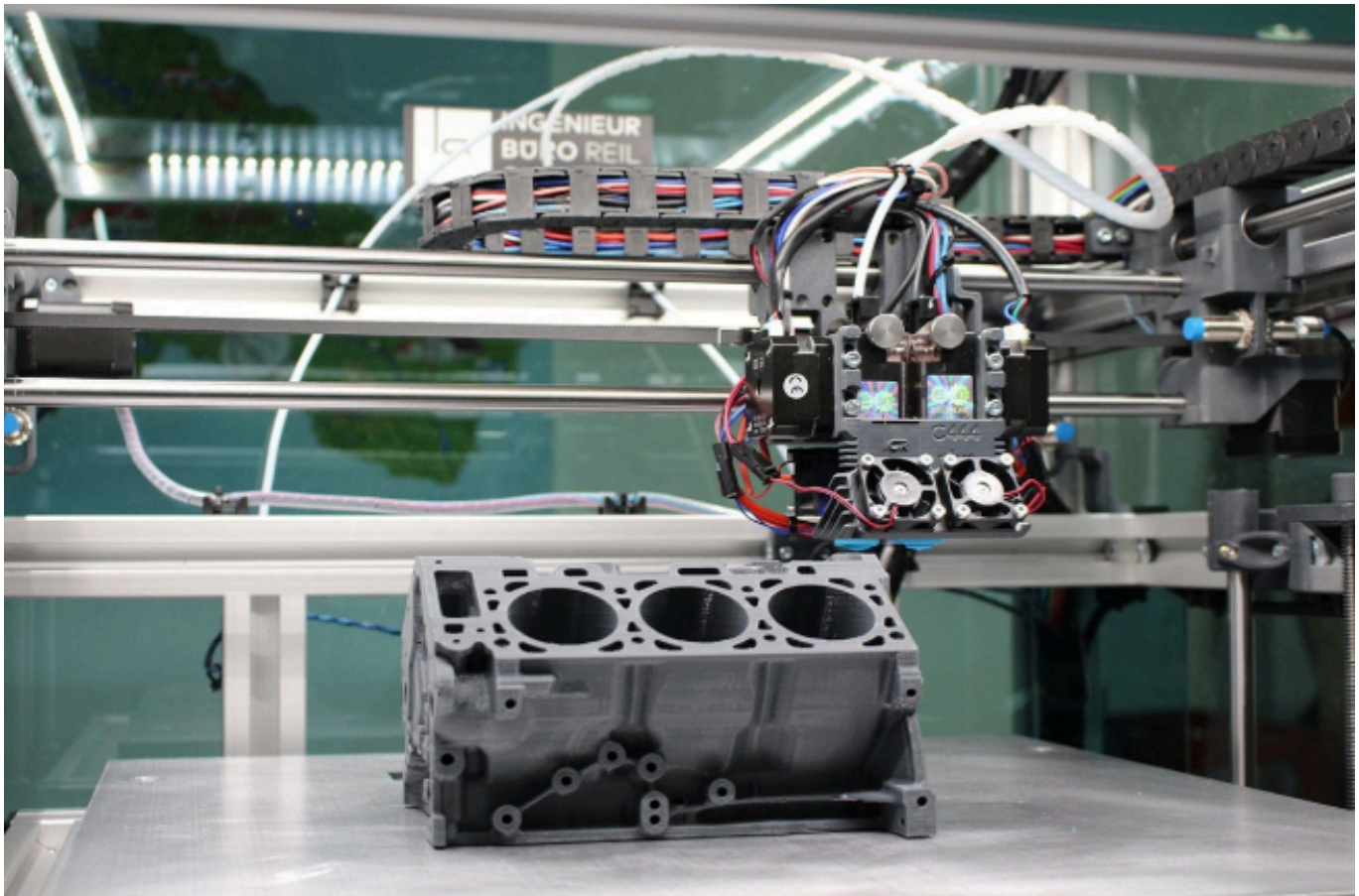


Qualification Pack



Electro - Mechanical Technician

QP Code: TNSD/CSC/Q0301

Version: 1.0

NSQF Level: 4

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Qualification Pack

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Qualification Pack

TNSD/CSC/Q0301: Electro - Mechanical Technician

Brief Job Description

Electro-Mechanical Technician job role is designed with various modules to equip learners with the necessary skills for diverse industrial applications. This qualification ultimately helps learners in the following ways: To become proficient in the design and manufacturing of control panels, industrial automation, and machine maintenance.

Personal Attributes

Design and construct a Mechatronics subsystem, write a PLC program to automate the system, and test its function for prescribed outputs. To secure employment opportunities across a wide range of sectors, including Engineering/Manufacturing industries, process industries, and the automobile industry. To pursue entrepreneurial ventures.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [TNSD/CSC/N0312: Apply Sensor Technology Principles.](#)
2. [TNSD/CSC/N0313: Operate and Troubleshoot Pneumatic and Electro-Pneumatic Systems.](#)
3. [TNSD/CSC/N0314: Operate and Troubleshoot Hydraulic and Electro-Hydraulic Systems.](#)
4. [TNSD/CSC/N0315: Program and Interface Microcontrollers with C](#)
5. [TNSD/CSC/N0316: Program and Troubleshoot Programmable Logic Controllers \(PLCs\).](#)
6. [TNSD/CSC/N0318: Operate and Program CNC Machines.](#)
7. [TNSD/CSC/N0317: Implement and Analyze Control Systems & Instrumentation.](#)
8. [TNSD/CSC/N0319: Design and Optimize Mechatronics Systems.](#)
9. [TNSD/CSC/N0320: Create 2D and 3D Drawings using SolidWorks.](#)
10. [TNSD/CSC/N0321: Simulate CNC Programs using Keller Software.](#)
11. [TNSD/CSC/N0322: Apply Technical Mathematics for Engineering Applications.](#)
12. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Qualification Pack

Sector	Capital Goods
Sub-Sector	
Occupation	Electro - Mechanical Technician
Country	India
NSQF Level	4
Credits	40
Aligned to NCO/ISCO/ISIC Code	3113.0401
Minimum Educational Qualification & Experience	Previous relevant Qualification of NSQF Level (Manufacturing Assembler - Electrical & Electronics (level 3.5))
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	16 Years
Last Reviewed On	NA
Next Review Date	07/10/2028
NSQC Approval Date	07/10/2025
Version	1.0
Reference code on NQR	QG-04-CG-04452-2025-V1-TNSDC
NQR Version	1.0

Qualification Pack

TNSD/CSC/N0312: Apply Sensor Technology Principles.

Description

Able to familiarize the sensors and their functionalities. Able to understand their applications in industries.

Scope

The scope covers the following :

- Understand the basics of different types and working of Sensors & Transducers.
- Familiarize with the Classification, characteristics & application of sensors. Familiarize with the Classification, Application of Proximity Sensors
- Understand different types of sensors used for Distance, Displacement, Force Temperature and Pressure

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Familiarize the basics of different types and working of Sensors & Transducers with the Classification , characteristics & application
- PC2.** Familiarize with the Classification , Application of Proximity Sensors
- PC3.** Familiarize with the Classification , Application of Temperature Sensors
- PC4.** Familiarize different types of sensors used for Distance & Displacement
- PC5.** Familiarize different types of sensors used for Force and Pressure
- PC6.** Familiarize how to select sensors for different Mechatronics application
- PC7.** Demonstrate functions of different sensors.
- PC8.** Demonstrate functions of different Transducers

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Familiarize the basics of different types and working of Sensors & Transducers with the Classification , characteristics & application
- KU2.** Familiarize with the Classification , Application of Proximity Sensors

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Familiarize different types of sensors used for Force and Pressure
- GS2.** Familiarize how to select sensors for different Mechatronics application

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	50	40	-	10
PC1. Familiarize the basics of different types and working of Sensors & Transducers with the Classification , characteristics & application	10	-	-	-
PC2. Familiarize with the Classification , Application of Proximity Sensors	8	-	-	-
PC3. Familiarize with the Classification , Application of Temperature Sensors	8	-	-	-
PC4. Familiarize different types of sensors used for Distance & Displacement	8	-	-	-
PC5. Familiarize different types of sensors used for Force and Pressure	8	-	-	-
PC6. Familiarize how to select sensors for different Mechatronics application	8	-	-	-
PC7. Demonstrate functions of different sensors.	-	20	-	5
PC8. Demonstrate functions of different Transducers	-	20	-	5
NOS Total	50	40	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0312
NOS Name	Apply Sensor Technology Principles.
Sector	Capital Goods
Sub-Sector	
Occupation	Electro - Mechanical Technician
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

TNSD/CSC/N0313: Operate and Troubleshoot Pneumatic and Electro-Pneumatic Systems.

Description

Able to understand the functioning of all the pneumatic components Able to make pneumatic circuits for industrial applications and troubleshoot.

Scope

The scope covers the following :

- Understand the Importance and basic concepts of Pneumatic Systems.
- Understand the function of Compressor and properties of compressed air.
- Familiarize the Symbols and schematic design of Pneumatic Circuits.
- Familiarize the function of Direction, Pressure, Flow control Valves.
- Understand different types of Actuators and Output devices.
- Understand Safety awareness, Testing and Trouble shooting of Pneumatic Systems.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Understand the Basic concepts of Pneumatic Systems, Properties of compressed air.

PC2. Understand the Function of Pneumatic Components and Compressor.

PC3. Familiarize the Symbols and schematics of Pneumatic Circuits.

PC4. Familiarize the function of Direction control Valves, Pressure control Valves , Flow control Valves.

PC5. Understand the different types of Actuators and Output devices.

PC6. Understand the Safety awareness, Testing and Trouble shooting of Pneumatic Systems.

PC7. Demonstrate Various components of pneumatics system and construct pneumatic circuit to check functionality.

PC8. Demonstrate the Construction of an electro-pneumatic circuit and check functionality of a process.

PC9. Perform Simulation of the Pneumatic circuit on simulation software and detect fault as per diagnostic procedure for Pneumatics system design.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understand the Basic concepts of Pneumatic Systems, Properties of compressed air.

KU2. Understand the Function of Pneumatic Components and Compressor.

KU3. Understand the different types of Actuators and Output devices.

Generic Skills (GS)

Logo
Not
Available



Qualification Pack

User/individual on the job needs to know how to:

- GS1.** Familiarize the Symbols and schematics of Pneumatic Circuits.
- GS2.** Demonstrate the Construction of an electro-pneumatic circuit and check functionality of a process.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	50	40	-	10
PC1. Understand the Basic concepts of Pneumatic Systems, Properties of compressed air.	5	-	-	-
PC2. Understand the Function of Pneumatic Components and Compressor.	10	-	-	-
PC3. Familiarize the Symbols and schematics of Pneumatic Circuits.	10	-	-	-
PC4. Familiarize the function of Direction control Valves, Pressure control Valves , Flow control Valves.	10	-	-	-
PC5. Understand the different types of Actuators and Output devices.	10	-	-	-
PC6. Understand the Safety awareness, Testing and Trouble shooting of Pneumatic Systems.	5	-	-	-
PC7. Demonstrate Various components of pneumatics system and construct pneumatic circuit to check functionality.	-	15	-	5
PC8. Demonstrate the Construction of an electro-pneumatic circuit and check functionality of a process.	-	15	-	5
PC9. Perform Simulation of the Pneumatic circuit on simulation software and detect fault as per diagnostic procedure for Pneumatics system design.	-	10	-	-
NOS Total	50	40	-	10

Logo
Not
Available



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0313
NOS Name	Operate and Troubleshoot Pneumatic and Electro-Pneumatic Systems.
Sector	Capital Goods
Sub-Sector	
Occupation	Electro - Mechanical Technician
NSQF Level	4
Credits	4
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

TNSD/CSC/N0314: Operate and Troubleshoot Hydraulic and Electro-Hydraulic Systems.

Description

Able to understand the functioning of all the hydraulic components Able to construct & execute hydraulic circuits for given industrial applications.

Scope

The scope covers the following :

- Understand the properties of hydraulic fluid and function of Reservoir
- Understand the Symbols and schematics and design of Hydraulic Circuits
- Familiarize the function of Direction, Pressure, Flow control Valves.
- Understand different types of Actuators and Output devices.
- Understand the various hydraulic oil types and their selection methods.
- Learn about Safety awareness, Testing and Trouble shooting of Hydraulic Systems

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Understand the properties of hydraulic fluid and function of Reservoir
- PC2.** Understand Function of Direction, Pressure, Flow control Valves.
- PC3.** Understand the different types of Actuators and Output devices.
- PC4.** Understand the types of hydraulic oils and selection
- PC5.** Understand the simple hydraulic circuits
- PC6.** Demonstrate installation of accessories in hydraulic system and troubleshoot defects
- PC7.** Construct hydraulic circuit for verification of various processes to assess functioning of valves and auxiliaries.
- PC8.** Demonstrate to Install hydraulic pump, motors and carryout maintenance of these components
- PC9.** Perform different hydraulic system such as Clamp control, injection control, reciprocating screw, oil filtration, hydraulic press control, accumulator control
- PC10.** Perform Simulation of the Hydraulic circuit on simulation software and detect fault as per diagnostic procedure for Hydraulics system design.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the properties of hydraulic fluid and function of Reservoir
- KU2.** Understand the different types of Actuators and Output devices.

Generic Skills (GS)

Logo
Not
Available



Qualification Pack

User/individual on the job needs to know how to:

GS1. Demonstrate installation of accessories in hydraulic system and troubleshoot defects

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	50	40	-	10
PC1. Understand the properties of hydraulic fluid and function of Reservoir	4	-	-	-
PC2. Understand Function of Direction, Pressure, Flow control Valves.	12	-	-	-
PC3. Understand the different types of Actuators and Output devices.	10	-	-	-
PC4. Understand the types of hydraulic oils and selection	12	-	-	-
PC5. Understand the simple hydraulic circuits	12	-	-	-
PC6. Demonstrate installation of accessories in hydraulic system and troubleshoot defects	-	8	-	2
PC7. Construct hydraulic circuit for verification of various processes to assess functioning of valves and auxiliaries.	-	8	-	2
PC8. Demonstrate to Install hydraulic pump, motors and carryout maintenance of these components	-	8	-	2
PC9. Perform different hydraulic system such as Clamp control, injection control, reciprocating screw, oil filtration, hydraulic press control, accumulator control	-	8	-	2
PC10. Perform Simulation of the Hydraulic circuit on simulation software and detect fault as per diagnostic procedure for Hydraulics system design.	-	8	-	2
NOS Total	50	40	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0314
NOS Name	Operate and Troubleshoot Hydraulic and Electro-Hydraulic Systems.
Sector	Capital Goods
Sub-Sector	
Occupation	Electro - Mechanical Technician, Electro - Mechanical Technician
NSQF Level	4
Credits	4
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

TNSD/CSC/N0315: Program and Interface Microcontrollers with C

Description

Able to familiarize the microcontroller construction & working. Able to create program in a microcontroller

Scope

The scope covers the following :

- Understand of the fundamentals of processor & controller.
- Familiarize Arduino board, programming language and instruction sets.
- Understand interfacing techniques of sensors and peripherals.
- Understand various peripheral communication techniques.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Understand the Fundamentals of processor & controller
- PC2.** Understand the Arduino board and programming language instruction sets
- PC3.** Familiarize pin configuration of Arduino board
- PC4.** Understand the Interfacing techniques of sensors and peripherals
- PC5.** Understand communication techniques of controllers
- PC6.** Familiarize the peripheral communication
- PC7.** Demonstrate an assembly language programme and interface peripherals to Microcontroller to check functioning.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the Fundamentals of processor & controller
- KU2.** Understand the Arduino board and programming language instruction sets

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Familiarize the peripheral communication

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	50	40	-	10
PC1. Understand the Fundamentals of processor & controller	8	-	-	-
PC2. Understand the Arduino board and programming language instruction sets	8	-	-	-
PC3. Familiarize pin configuration of Arduino board	8	-	-	-
PC4. Understand the Interfacing techniques of sensors and peripherals	8	-	-	-
PC5. Understand communication techniques of controllers	8	-	-	-
PC6. Familiarize the peripheral communication	10	-	-	-
PC7. Demonstrate an assembly language programme and interface peripherals to Microcontroller to check functioning.	-	40	-	10
NOS Total	50	40	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0315
NOS Name	Program and Interface Microcontrollers with C
Sector	Capital Goods
Sub-Sector	
Occupation	Electro - Mechanical Technician
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

TNSD/CSC/N0316: Program and Troubleshoot Programmable Logic Controllers (PLCs).

Description

Able to familiarize the construction and working of a PLC. Able to create a program in PLC for given application.

Scope

The scope covers the following :

- Understand the fundamentals of programmable logic controllers.
- Familiarization of PLC programming languages.
- Understanding the application of PLC with timers and counters.
- Understanding the testing and troubleshooting techniques in PLC.
- Understand the usage of different programming languages.
- Familiarization of Advanced PLC Functions in line with industrial requirement.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Understand the Fundamentals of programmable logic controllers

PC2. Familiarize the PLC programming languages

PC3. Understand the Application of PLC with timers and counters

PC4. Understand the testing and troubleshooting techniques in PLC

PC5. Understand the simple PLC programs

PC6. Understand the application requirement before programming

PC7. Familiarize Advanced PLC Functions in line with industrial requirement

PC8. Perform programming of PLC and interfacing with other devices.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understand the testing and troubleshooting techniques in PLC

KU2. Understand the application requirement before programming

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Familiarize Advanced PLC Functions in line with industrial requirement

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	50	40	-	10
PC1. Understand the Fundamentals of programmable logic controllers	5	-	-	-
PC2. Familiarize the PLC programming languages	10	-	-	-
PC3. Understand the Application of PLC with timers and counters	5	-	-	-
PC4. Understand the testing and troubleshooting techniques in PLC	5	-	-	-
PC5. Understand the simple PLC programs	10	-	-	-
PC6. Understand the application requirement before programming	10	-	-	-
PC7. Familiarize Advanced PLC Functions in line with industrial requirement	5	-	-	-
PC8. Perform programming of PLC and interfacing with other devices.	-	40	-	10
NOS Total	50	40	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0316
NOS Name	Program and Troubleshoot Programmable Logic Controllers (PLCs).
Sector	Capital Goods
Sub-Sector	
Occupation	Electro - Mechanical Technician
NSQF Level	4
Credits	4
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

TNSD/CSC/N0318: Operate and Program CNC Machines.

Description

Able to understand the CNC programming for machining a given part on CNC milling/CNC lathe.

Scope

The scope covers the following :

- Getting Introduced to the Basic Concepts.
- Understand the constructional features of CNC machines.
- To familiarize with concept of offset and different modes of operation.
- To familiarize with tools & tool holders

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Getting Introduced to the Basic Concepts

PC2. Understand the constructional features of CNC machines

PC3. Understand the coordinate system & codes

PC4. Familiarize with concept of offset and different modes of operation

PC5. Familiarize with Programming

PC6. Familiarize with tools & tool holders

PC7. Demonstrate to Set (both job and tool) CNC turn centre and milling machine to produce simple components as per drawing

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understand the constructional features of CNC machines

KU2. Understand the coordinate system & codes

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Familiarize with concept of offset and different modes of operation

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	50	40	-	10
PC1. Getting Introduced to the Basic Concepts	2	-	-	-
PC2. Understand the constructional features of CNC machines	4	-	-	-
PC3. Understand the coordinate system & codes	4	-	-	-
PC4. Familiarize with concept of offset and different modes of operation	5	5	-	-
PC5. Familiarize with Programming	10	5	-	-
PC6. Familiarize with tools & tool holders	10	5	-	-
PC7. Demonstrate to Set (both job and tool) CNC turn centre and milling machine to produce simple components as per drawing	15	25	-	10
NOS Total	50	40	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0318
NOS Name	Operate and Program CNC Machines.
Sector	Capital Goods
Sub-Sector	
Occupation	Electro - Mechanical Technician, Electro - Mechanical Technician
NSQF Level	4
Credits	4
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

TNSD/CSC/N0317: Implement and Analyze Control Systems & Instrumentation.

Description

Able to understand the concept of control system for creating a stable Mechatronics subsystem.

Scope

The scope covers the following :

- Understand the concepts of control systems
- Familiarize the graphical representation of the systems
- Familiarize the control system components
- Understand the time and frequency response
- Understand the roots location of a system
- Understand the stability of systems
- Familiarize the electrical measuring instruments

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Understand the concepts of control systems
- PC2.** Familiarize the graphical representation of the systems
- PC3.** Familiarize the control system components
- PC4.** Understand & analyze the time and frequency response
- PC5.** Understand the roots location of a system
- PC6.** Understand the stability of systems
- PC7.** Familiarize the electrical measuring instruments
- PC8.** Demonstrate basic concepts and principles of system control to perform practical operations.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the concepts of control systems
- KU2.** Understand & analyze the time and frequency response

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Familiarize the control system components

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	50	40	-	10
PC1. Understand the concepts of control systems	5	-	-	-
PC2. Familiarize the graphical representation of the systems	8	-	-	-
PC3. Familiarize the control system components	8	-	-	-
PC4. Understand & analyze the time and frequency response	8	-	-	-
PC5. Understand the roots location of a system	8	-	-	-
PC6. Understand the stability of systems	8	-	-	-
PC7. Familiarize the electrical measuring instruments	5	-	-	-
PC8. Demonstrate basic concepts and principles of system control to perform practical operations.	-	40	-	10
NOS Total	50	40	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0317
NOS Name	Implement and Analyze Control Systems & Instrumentation.
Sector	Capital Goods
Sub-Sector	
Occupation	Electro - Mechanical Technician
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

TNSD/CSC/N0319: Design and Optimize Mechatronics Systems.

Description

Able to understand the functioning of a mechatronics system. Able to create a mechatronics system by using PLC, HMI, Drives and MPS stations.

Scope

The scope covers the following :

- Understand the importance and basic concepts of Mechatronics system
- Understand the working of guide ways and various drive systems used in the Mechatronics system
- Understand the control architecture of a Mechatronics system
- Understand the working of indexing mechanisms used in the Mechatronics system
- Understand the key elements for designing the Mechatronics system

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Understand the basic concepts of Mechatronics system
- PC2.** Understand the working of guide ways in the Mechatronics system
- PC3.** Understand the need and different methods of control system
- PC4.** Identify the inputs & outputs of any given Mechatronics system and plan for communication channel
- PC5.** Understand the control architecture of given Mechatronics system and identify its components.
- PC6.** Demonstrate the working of indexing mechanisms used in the Mechatronics system and create parts for mechatronic subsystem
- PC7.** Demonstrate the designing of a Mechatronics system and perform the programming in PLC.
- PC8.** Perform the selection of proper electrical drive system for a mechatronic subsystem.
- PC9.** Demonstrate the analysis of existing Mechatronics system and implement methods to improve its efficiency.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the working of guide ways in the Mechatronics system
- KU2.** Understand the need and different methods of control system

Generic Skills (GS)

User/individual on the job needs to know how to:

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Not
Available



Qualification Pack

- GS1.** Identify the inputs & outputs of any given Mechatronics system and plan for communication channel
- GS2.** Demonstrate the designing of a Mechatronics system and perform the programming in PLC.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	50	40	-	10
PC1. Understand the basic concepts of Mechatronics system	5	-	-	-
PC2. Understand the working of guide ways in the Mechatronics system	5	-	-	-
PC3. Understand the need and different methods of control system	5	-	-	-
PC4. Identify the inputs & outputs of any given Mechatronics system and plan for communication channel	5	-	-	-
PC5. Understand the control architecture of given Mechatronics system and identify its components.	5	5	-	-
PC6. Demonstrate the working of indexing mechanisms used in the Mechatronics system and create parts for mechatronic subsystem	5	10	-	3
PC7. Demonstrate the designing of a Mechatronics system and perform the programming in PLC.	6	10	-	2
PC8. Perform the selection of proper electrical drive system for a mechatronic subsystem.	6	10	-	2
PC9. Demonstrate the analysis of existing Mechatronics system and implement methods to improve its efficiency.	8	5	-	3
NOS Total	50	40	-	10

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0319
NOS Name	Design and Optimize Mechatronics Systems.
Sector	Capital Goods
Sub-Sector	
Occupation	Electro - Mechanical Technician, Electro - Mechanical Technician
NSQF Level	4
Credits	5
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

TNSD/CSC/N0320: Create 2D and 3D Drawings using SolidWorks.

Description

Able to draw 2D,3D drawing using Solid works software.

Scope

The scope covers the following :

- Understanding Basics of 2D and 3D drawing

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Drawing 2D profiles and Modelling.

PC2. 3D modelling of objects by using different tools.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Drawing 2D profiles and Modelling.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Three D modelling of objects by using different tools.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	80	-	20
PC1. Drawing 2D profiles and Modelling.	-	40	-	10
PC2. 3D modelling of objects by using different tools.	-	40	-	10
NOS Total	-	80	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0320
NOS Name	Create 2D and 3D Drawings using SolidWorks.
Sector	Capital Goods
Sub-Sector	
Occupation	Electro - Mechanical Technician, Electro - Mechanical Technician
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

TNSD/CSC/N0321: Simulate CNC Programs using Keller Software.

Description

Simulation of CNC program by Keller software.

Scope

The scope covers the following :

- To get familiar with different tools and functions of KELLER software

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Familiarization of basic programming, Feed & Speed , Cycle time, cutting tools etc. of CNC lathe
- PC2.** Familiarization with parts and understanding the working environment of CNC Milling machine
- PC3.** Familiarization of basic programming, Feed & Speed , Cycle time, cutting tools etc. of CNC Milling machine
- PC4.** Familiarization of ISO cycles and its implementation in CNC Lathe
- PC5.** Familiarization of various control system of CNC Lathe
- PC6.** Conversion of raw material to finished part using CNC Lathe
- PC7.** Familiarization of ISO cycles and it's implementation in CNC Milling Machine
- PC8.** Familiarization of control system of CNC Milling
- PC9.** Conversion of raw material to finished part using CNC Milling

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Familiarization of ISO cycles and its implementation in CNC Lathe
- KU2.** Familiarization of basic programming, Feed & Speed , Cycle time, cutting tools etc. of CNC lathe

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Familiarization of ISO cycles and it's implementation in CNC Milling Machine

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	80	-	20
PC1. Familiarization of basic programming, Feed & Speed , Cycle time, cutting tools etc. of CNC lathe	-	5	-	2
PC2. Familiarization with parts and understanding the working environment of CNC Milling machine	-	5	-	2
PC3. Familiarization of basic programming, Feed & Speed , Cycle time, cutting tools etc. of CNC Milling machine	-	10	-	2
PC4. Familiarization of ISO cycles and its implementation in CNC Lathe	-	10	-	2
PC5. Familiarization of various control system of CNC Lathe	-	10	-	2
PC6. Conversion of raw material to finished part using CNC Lathe	-	10	-	2
PC7. Familiarization of ISO cycles and it's implementation in CNC Milling Machine	-	10	-	2
PC8. Familiarization of control system of CNC Milling	-	10	-	3
PC9. Conversion of raw material to finished part using CNC Milling	-	10	-	3
NOS Total	-	80	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0321
NOS Name	Simulate CNC Programs using Keller Software.
Sector	Capital Goods
Sub-Sector	
Occupation	Electro - Mechanical Technician
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

TNSD/CSC/N0322: Apply Technical Mathematics for Engineering Applications.

Description

Understand the concept of basic mathematics used for doing calculations related to engineering application and general purpose.

Scope

The scope covers the following :

- Understand the basic Concepts of principles of analytical geometry, Line equations, slope, intercept, Areas of Triangles, Quadrilaterals, parallel and perpendicular lines, finding the distance, midpoints and its applications with related problems.
- Familiarize the differential and integral calculus, and problems related to their applications.
- Learn to find solution of triangles, sine rule, cosine rule, projection formula, trigonometric identities, height and distance problems, sine bar and taper related problems.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Learn the Concepts of fundamental Analytical geometry
- PC2.** Find the slope of line , parallel and perpendicular lines
- PC3.** Find the distance between the vertices Transversal of line
- PC4.** Calculate the area of triangles and quadrilaterals , mid points, Equation of lines and related problems
- PC5.** Calculate the Centroid , in center, Ortho center of a triangles and related problems.
- PC6.** Understand the basic concept & application of calculus
- PC7.** Solve the linear and non linear equations , Ordinary differential & partial differential equations.
- PC8.** Find the order and degree of the differential equations, solution of ODE& PDE.
- PC9.** Learn the Differential rule , Chain rule, Bernoulli rule and related problems
- PC10.** Learn Basic concept of principle of Trigonometry and its applications
- PC11.** Solve the problems using trigonometry application
- PC12.** Find the Solution of acute , obtuse angled triangles and related problems
- PC13.** Find the values of angles by using trigonometric table values, Angle value for the special angles, Solution of circumscribed triangles by sine rule and practice problems.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Learn the Concepts of fundamental Analytical geometry

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Qualification Pack

KU2. Understand the basic concept & application of calculus

KU3. Learn the Differential rule , Chain rule, Bernoulli rule and related problems

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Find the slope of line , parallel and perpendicular lines

GS2. Find the distance between the vertices Transversal of line

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	100	-	-	-
PC1. Learn the Concepts of fundamental Analytical geometry	5	-	-	-
PC2. Find the slope of line , parallel and perpendicular lines	5	-	-	-
PC3. Find the distance between the vertices Transversal of line	5	-	-	-
PC4. Calculate the area of triangles and quadrilaterals , mid points, Equation of lines and related problems	10	-	-	-
PC5. Calculate the Centroid , in center, Ortho center of a triangles and related problems.	10	-	-	-
PC6. Understand the basic concept & application of calculus	5	-	-	-
PC7. Solve the linear and non linear equations , Ordinary differential & partial differential equations.	10	-	-	-
PC8. Find the order and degree of the differential equations, solution of ODE& PDE.	10	-	-	-
PC9. Learn the Differential rule , Chain rule, Bernoulli rule and related problems	10	-	-	-
PC10. Learn Basic concept of principle of Trigonometry and its applications	5	-	-	-
PC11. Solve the problems using trigonometry application	10	-	-	-
PC12. Find the Solution of acute , obtuse angled triangles and related problems	10	-	-	-
PC13. Find the values of angles by using trigonometric table values, Angle value for the special angles, Solution of circumscribed triangles by sine rule and practice problems.	5	-	-	-

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Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	100	-	-	-

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Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0322
NOS Name	Apply Technical Mathematics for Engineering Applications.
Sector	Capital Goods
Sub-Sector	
Occupation	Electro - Mechanical Technician
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e-mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

Qualification Pack

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

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Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

<1. Assessment System Overview:

- Batches assigned to the assessment agencies for conducting the assessment on SIP or email
- Assessment agencies send the assessment confirmation to VTP/TC looping TNSDC
- Assessment agency deploys the ToA certified Assessor for executing the assessment
- TNSDC monitors the assessment process & records

2. Testing Environment:

- Check the Assessment location, date and time
- If the batch size is more than 30, then there should be 2 Assessors.
- Check that the allotted time to the candidates to complete Theory & Practical Assessment is correct.

3. Assessment Quality Assurance levels/Framework:

Qualification Pack

- Question bank is created by the Subject Matter Experts (SME) are verified by the other SME
- Questions are mapped to the specified assessment criteria
- Assessor must be ToA certified & trainer must be ToT Certified

4. Types of evidence or evidence-gathering protocol:

- Time-stamped & geotagged reporting of the assessor from assessment location
- Centre photographs with signboards and scheme specific branding

5. Method of verification or validation:

- Surprise visit to the assessment location

6. Method for assessment documentation, archiving, and access

- Hard copies of the documents are stored

On the Job:

1. Each module will be assessed separately.

2. The candidate must score 60% in each module to successfully complete the OJT.

3. Tools of Assessment that will be used for assessing whether the candidate is having desired skills and etiquette of dealing with customers, understanding needs & requirements, assessing the customer and perform Soft Skills effectively:

- Videos of Trainees during OJT

4. Assessment of each Module will ensure that the candidate is able to:

- Effective engagement with the customers
- Understand the working of various tools and equipment

Minimum Aggregate Passing % at QP Level : 60

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
TNSD/CSC/N0312.Apply Sensor Technology Principles.	50	40	0	10	100	10
TNSD/CSC/N0313.Operate and Troubleshoot Pneumatic and Electro-Pneumatic Systems.	50	40	0	10	100	10
TNSD/CSC/N0314.Operate and Troubleshoot Hydraulic and Electro-Hydraulic Systems.	50	40	0	10	100	10
TNSD/CSC/N0315.Program and Interface Microcontrollers with C	50	40	0	10	100	10
TNSD/CSC/N0316.Program and Troubleshoot Programmable Logic Controllers (PLCs).	50	40	0	10	100	10
TNSD/CSC/N0318.Operate and Program CNC Machines.	50	40	0	10	100	10
TNSD/CSC/N0317.Implement and Analyze Control Systems & Instrumentation.	50	40	0	10	100	10
TNSD/CSC/N0319.Design and Optimize Mechatronics Systems.	50	40	0	10	100	10
TNSD/CSC/N0320.Create 2D and 3D Drawings using SolidWorks.	0	80	0	20	100	5
TNSD/CSC/N0321.Simulate CNC Programs using Keller Software.	0	80	0	20	100	5
TNSD/CSC/N0322.Apply Technical Mathematics for Engineering Applications.	100	0	0	0	100	5
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	5
Total	520	510	-	120	1150	100

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Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.